

Cloudflare Radar

2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's global network in 2024.

17%

Global Traffic Growth

3.3×

Starlink Traffic Surge

225

Major Disruptions

6.5%

Traffic Mitigated

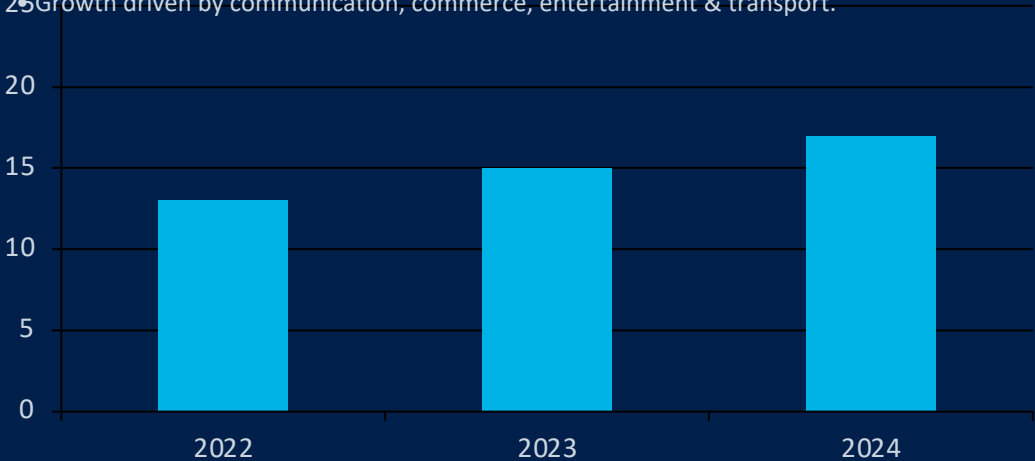
17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet Traffic Growth

+17%

Year-over-Year

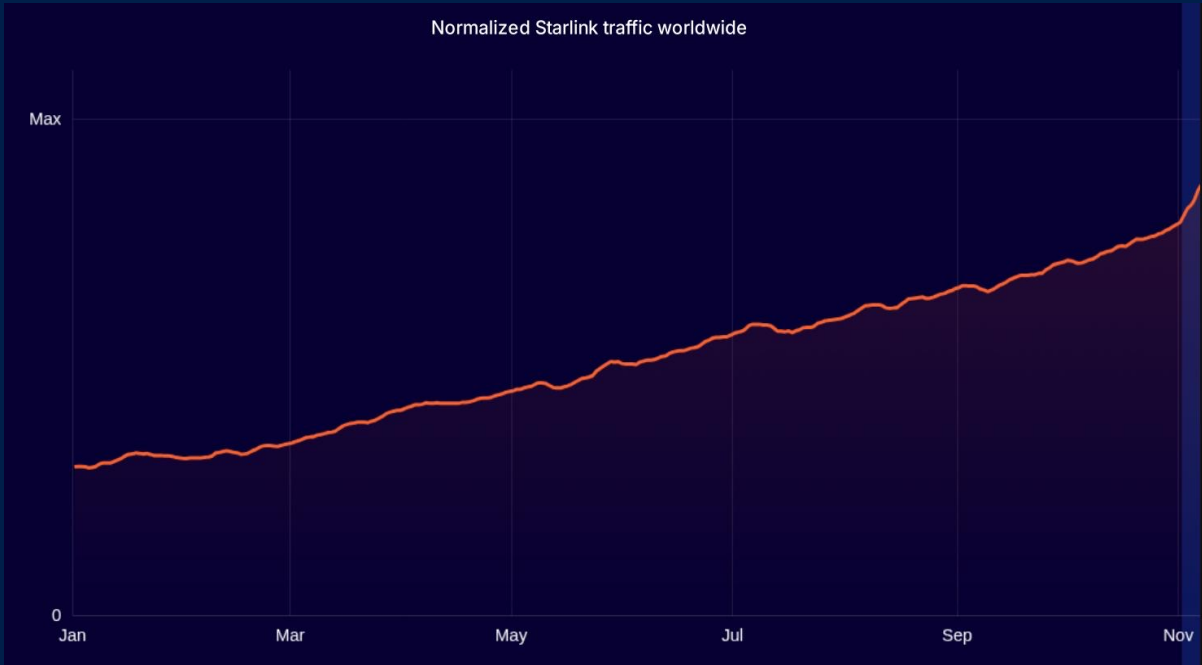
- Trend line starts mid-January to normalize for post-holiday activity.
- 25Growth driven by communication, commerce, entertainment & transport.



SpaceX Starlink Traffic

3.3x

Traffic Growth in 2024



- Satellite Internet now reaching previously underserved regions.
- Trend line scaled relative to peak observed request volume.

Google Leads Top 10 Services as Generative AI Rises

Top 10 Most Popular Internet Services (2024)

#1	Google	2nd consecutive year
#2	Facebook	
#3	Apple	
#4	TikTok	
#5	AWS	
#6	Microsoft	
#7	Instagram	
#8	YouTube	
#9	Amazon	
#10	WhatsApp	

Based on Cloudflare 1.1.1.1 DNS resolver aggregate data

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

The Rise of Generative AI

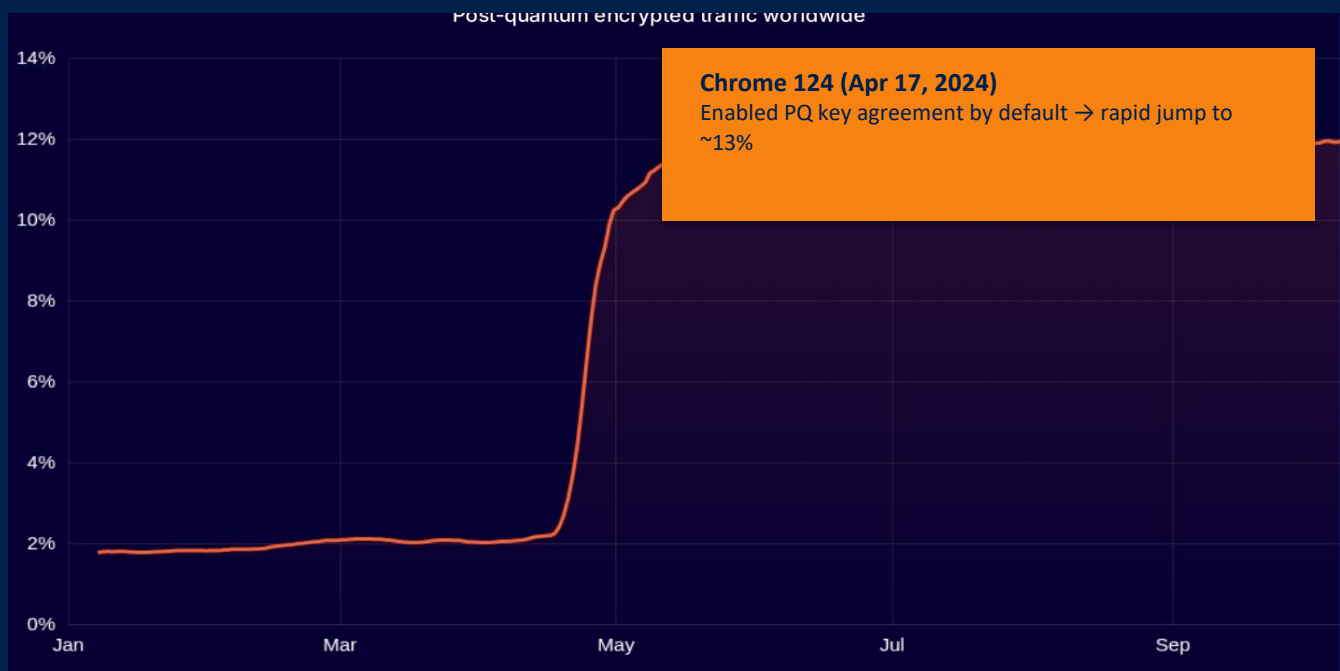
Generative AI services continued their rapid rise throughout 2024, representing one of the most significant shifts in Internet traffic patterns observed via Cloudflare's network.

The Metaverse category, by contrast, receded significantly from its peak prominence.

Ranking methodology: aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Post-Quantum Encryption Adoption (2024)



2% → 13%

Post-Quantum TLS 1.3

Dramatic adoption following Chrome 124 default enablement (Apr 2024). Firefox & Safari also rolling out.

21%

HTTP/3 Adoption

Runs on QUIC — faster connections, packet-loss mitigation, and encryption by default.

~12%

Go — Top API Language

33%

Traffic from iOS

41%

Mobile Device Traffic

89%

Search via Google

28%

IPv6 Dual-Stack

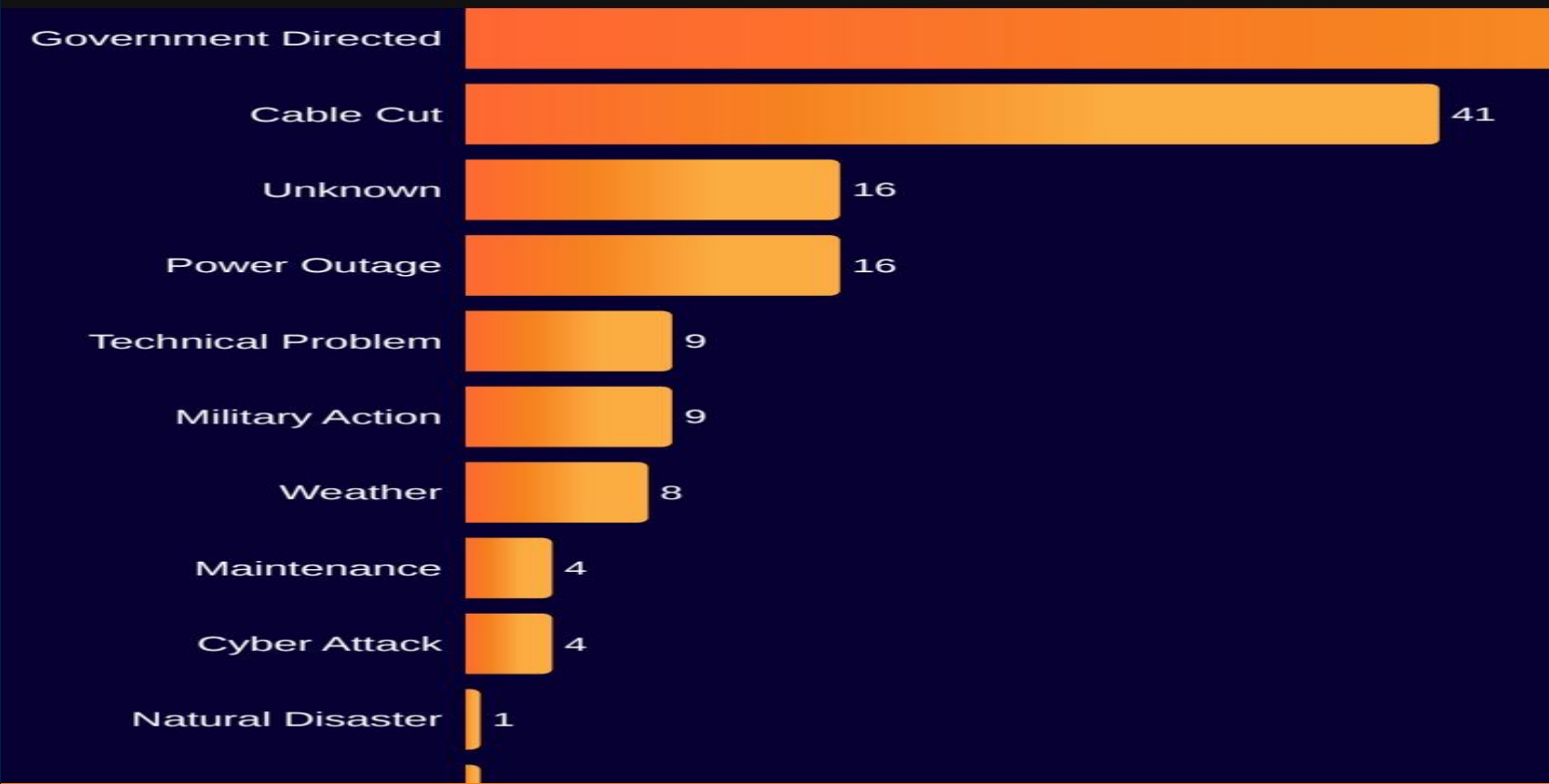
225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

Major Internet Disruptions
observed globally in 2024

CAUSE	COUNT
Government Directed	116
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

Internet Disruption Causes — 2024



Key Insight: Government-directed shutdowns (116+) account for more than half of all major disruptions — far exceeding cable cuts (41) and power outages (16).

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic Mitigated

Of all traffic processed by Cloudflare — indicates scale of ongoing threats.

68%

Bot Traffic from Top 10 Countries

Global bot traffic concentration among just 10 countries.

4.3%

Malicious Emails

Of all emails processed by Cloudflare email security.

43%

Deceptive Links

Of malicious emails contained a deceptive link to trick recipients.

Most Attacked Sector

Gambling & Games

The Gambling and Games sector received the highest volume of cyber attacks mitigated by Cloudflare in 2024.

RPKI valid routes increased by 6.4%, improving Internet routing security.

Log4j Persistence

CVE-2021-44228

First disclosed: December 2021

Log4j exploitation continues to be actively targeted by attackers more than two years after initial disclosure — underscoring the persistent nature of widely known, unpatched vulnerabilities.

Critical lesson: legacy vulnerability hygiene remains essential.

Email Security Threats

4.3%

of emails were malicious

43%

contained deceptive links

.bar

top-level domain with most malicious/spam email

Five Takeaways Shaping the 2025 Internet Outlook

01

Traffic Growth Trajectory Continues

Global Internet traffic grew 17% in 2024, with Starlink's 3.3× surge accelerating connectivity expansion into underserved regions. Network infrastructure must scale to accommodate sustained double-digit annual growth.

02

Post-Quantum Encryption Enters Mainstream

The jump from ~2% to 13% of TLS 1.3 traffic in a single year — triggered by Chrome 124's default enablement — signals that post-quantum cryptography is no longer theoretical. Organizations should begin migration planning now.

03

Connectivity Fragility from Government Actions

Government-directed shutdowns caused 116+ of 225 major disruptions in 2024 — more than half. Resilience strategies must account for deliberate, politically motivated outages alongside technical failures.

04

Persistent Security Threats Demand Continuous Vigilance

6.5% of traffic was mitigated by Cloudflare; Log4j exploitation persists 3+ years post-disclosure. The Gambling/Games sector remains a prime target. Zero-trust and automated mitigation are non-negotiable baselines.

05

Evolving Developer & Protocol Ecosystem

Go leads API client languages (~12%); HTTP/3 adoption reached 21%. The developer stack is shifting — organizations should align API gateway strategies and CDN configurations to modern protocol defaults.

Explore the full interactive report at radar.cloudflare.com/year-in-review/2024